Luminary Films: Quantified Risk Analysis of a Lost Shoot Day

Fault tree, event trees and influence diagram for a Scottish film production company running 60 to 90 shoot days a year.

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THE QUESTION

Luminary Films depends on one local equipment supplier with no formal contract. How much does that concentration actually cost, and which control is worth buying?

1 The exposure is concentrated in one supplier relationship

The fault tree estimates a 4.32% probability of losing a scheduled shoot day, roughly 3.24 losses in a central 75-day year before controls. Supplier same-day cancellation accounts for 34% of full-loss causes, and CastleLight is the only local supplier able to respond within ninety minutes.

2 Two independent methods were built, and they agreed

The influence diagram reproduced the fault tree's baseline full-loss probability under the same central inputs, then went further by ranking eight control portfolios by expected annual value. Agreement between two methods built independently is worth more than either result alone.

3 Only one control portfolio pays for itself

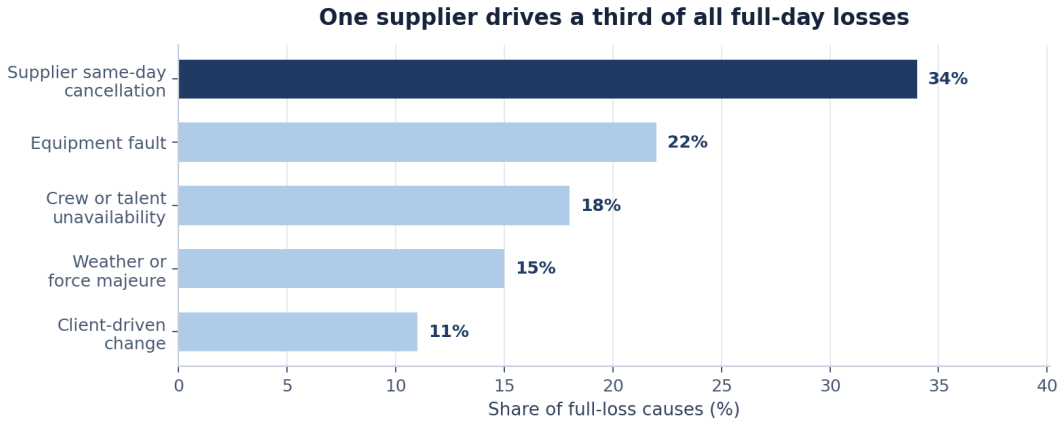
A stronger supplier service-level agreement (D1) returns £1,377 annual net benefit at a benefit-cost ratio of 1.31, and is the least-cost option in 96.3% of 5,000 uncertainty simulations. Buying backup equipment or adding a coordinator role destroys value at current assumptions.

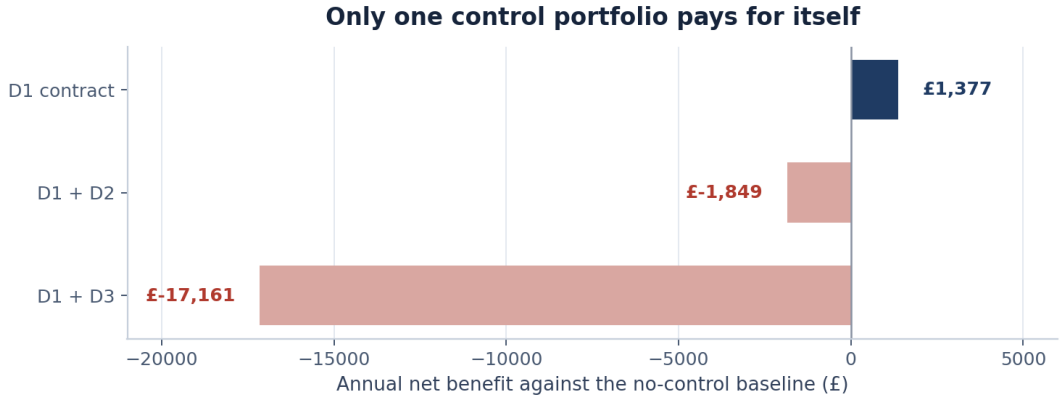
4 The recommendation carries its own break-even conditions

Negotiate D1 with enforceable continuity, step-in and break provisions at a premium below £5,877. Hold the equipment purchase as a conditional capital option, triggered only by a verified five-year economic life or roughly 90% effectiveness.

LIMITATIONS, AND WHAT WOULD CHANGE THE ANSWER

Control effectiveness figures are judgemental, not observed, and the report says they should be replaced with actual performance data once available. The supplier branch is anchored on a single cancellation in a reconstructed 67 bookings, which is a thin base for a probability that drives the whole result. The permit and access pathway stays qualitative because no reliable local frequency was found, so it sits in the bow-tie as a grey path rather than being assigned a number it does not have.





Built with Reliability Workbench · fault tree and event tree analysis · influence diagrams · bow-tie · 5,000-run Monte Carlo

Delivered quantified risk model · eight ranked control portfolios · cost-benefit analysis · negotiation position